Quiz 6: Solutions

Give exact answers and justify your reasoning. Do not use electronic tools. All functions are real-valued, and angles are measured in radians. You may use the definitions, limit laws and differentiation rules covered through Lecture 7.

1. Find and simplify the derivative of

$$F(x) = \frac{\sin x}{1 + \cos x}.$$

State the domains of F and F'.

Solution.

The denominator is zero exactly when $x = (2k+1)\pi$ for an integer k. Elsewhere the quotient rule gives

$$\begin{aligned} F'(x) &= \frac{\cos x(1 + \cos x) - \sin x(-\sin x)}{(1 + \cos x)^2} \\ &= \frac{\cos x + \cos^2 x + \sin^2 x}{(1 + \cos x)^2} = \boxed{\frac{1}{1 + \cos x}}. \end{aligned}$$

Both F and F' have domain

$$\boxed{\mathbb{R} \setminus \{(2k + 1)\pi : k \in \mathbb{Z}\}.}$$

2. Let $f(x) = x|x|$. Find $f'(x)$ for $x \neq 0$, then use the definition to determine whether $f'(0)$ exists. Is f' continuous at 0?

Solution.

For $x > 0$, $f(x) = x^2$, so $f'(x) = 2x$. For $x < 0$, $f(x) = -x^2$, so $f'(x) = -2x$. At zero, $f(0) = 0$, and for $h \neq 0$,

$$\frac{f(h) - f(0)}{h} = \frac{h|h|}{h} = |h| \longrightarrow 0.$$

Hence $f'(0)$ exists and equals 0. Combining the cases,

$$\boxed{f'(x) = 2|x| \qquad (x \in \mathbb{R}).}$$

Since $|f'(x) - f'(0)| = 2|x| \to 0$ as $x \to 0$, the derivative f' is continuous at zero.